# Telehealth Hypertension Analytics

Explained, audited blood pressure staging from 24-hour monitoring

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## 1 Project Description

**What it is.** A decision-support tool that reads a patient's full 24-hour blood pressure record, suggests a hypertension stage, shows which measurements drove the result, and flags uncertain cases for a clinician.

**Who it helps.** Clinicians who review ambulatory monitoring (ABPM), where a wearable cuff takes about 48 readings across a day and night.

**1 — Record in**
48 readings from one 24-hour cuff recording

**2 — Stage + explain**
Normal, Elevated, Stage 1 or Stage 2, with reasons

**3 — Clinician decides**
Low-confidence cases are flagged for review

**What was built.** A machine learning model, a secured web service (API), a dashboard page and 44 automated tests, released together as one tagged version on GitHub.

**Data.** Real patient records are protected, so every result here uses 1,800 simulated patients.

## 2 Project Justification

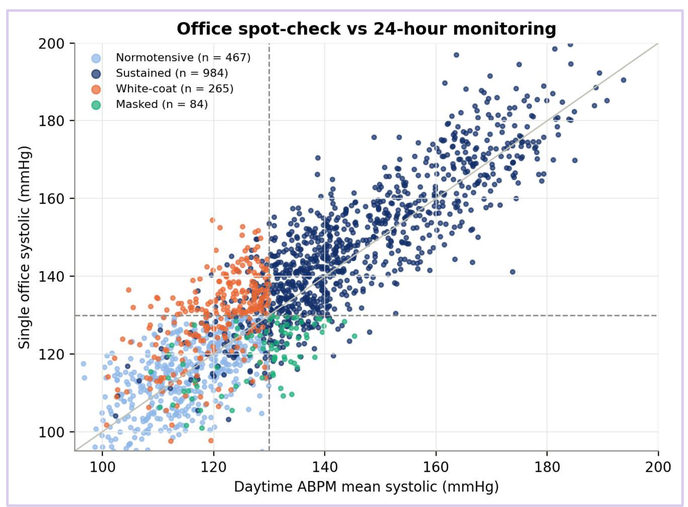
High blood pressure is a leading modifiable risk factor for cardiovascular disease, yet it is usually diagnosed from a few office readings (Whelton et al., 2018).

A single reading can mislead in both directions: white-coat hypertension looks high only in the clinic, and masked hypertension looks normal in the clinic but is high outside it.

**25.5%** one office reading disagrees with true status

**54%** office-rule staging accuracy

**48** readings per patient to review by hand




*Each dot is a simulated patient. Orange: high in clinic but not hypertensive (white-coat). Green: normal in clinic but hypertensive (masked).*

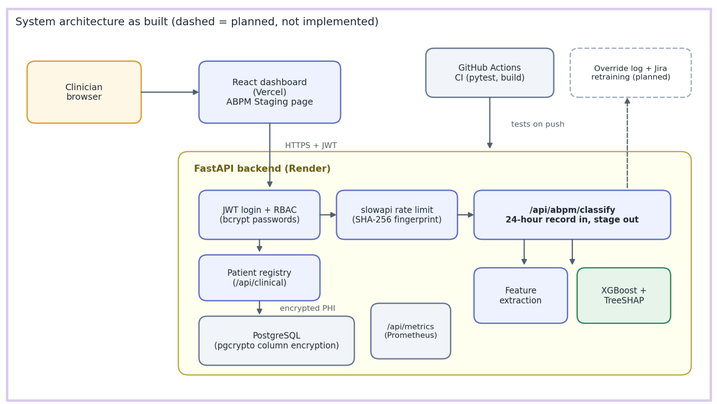
## 3 Functional Requirements, Specifications and Standards

| Requirement | Specification as built | Standard or evidence |
|---|---|---|
| **Stage a patient from a 24-hour record** | One API call takes about 48 readings and returns Normal, Elevated, Stage 1 or Stage 2 | 2017 ACC/AHA staging (Whelton et al., 2018) |
| **Explain every result** | Top four contributing measurements plus a plain-English summary | TreeSHAP attributions (Lundberg et al., 2020) |
| **Give trustworthy confidence** | Calibrated probabilities (error 0.014); results under 65% flagged for review | Reporting guided by TRIPOD+AI (Collins et al., 2024) |
| **Respond within 200 ms** | Measured 22 ms for prediction plus explanation | Project service-level objective |
| **Protect patient data** | Hashed passwords, role-based access tokens, rate limiting, encrypted patient columns | HIPAA Security Rule safeguards (U.S. Department of Health and Human Services, 2013) |
| **Be reproducible and tested** | Fixed random seeds, 44 automated tests, tagged release | Documented in a model card (Mitchell et al., 2019) |

## 4 Design Process

Six stages, reshaped by two reviews: professor feedback on holdout leakage, then a self-audit.

1. **Define requirements** — Clinical need, staging rules and security needs set first
2. **Simulate and prepare data** — 1,800 patients, 48 readings each, with noise, artifacts and gaps
3. **Select the model fairly** — Choices by cross-validation on training data only; 540 patients held back for one final test
4. **Build the service** — Secured API and dashboard page; model, model card and tests released together
5. **Audit the whole project** — Claims checked against the code; security and evaluation gaps found and fixed
6. **Report honestly** — Limits and open issues published with the results




*System as built. The dashed box (clinician override logging and retraining) is planned, not implemented.*

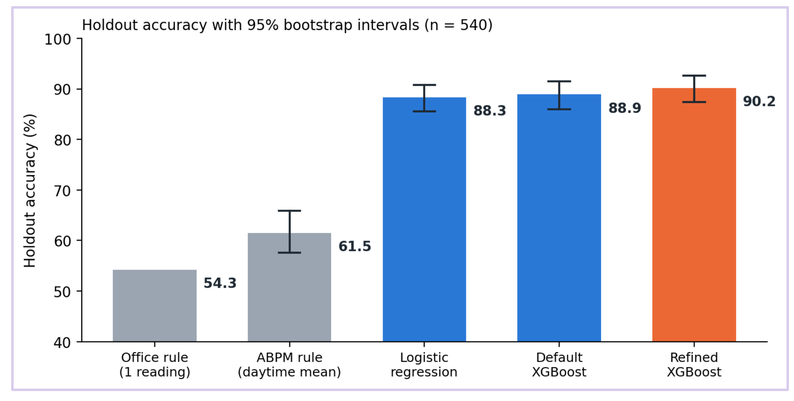
## 5 Relevant Data Summaries

**90.2%** of 540 unseen patients staged correctly

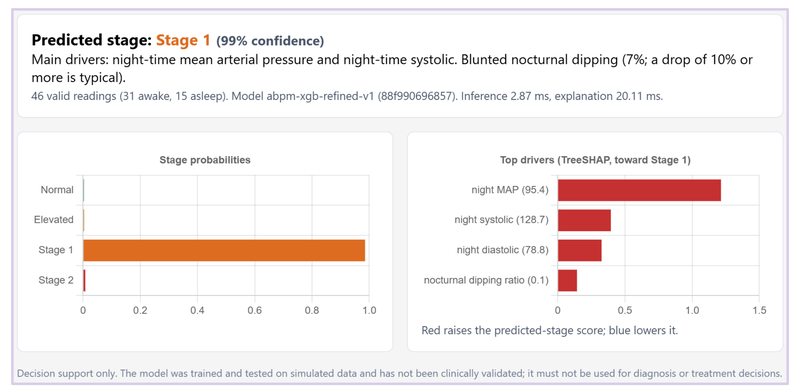
**97%** of Stage 2 patients caught (131 of 135)

**0** patients staged more than one step from the true stage

**22 ms** to predict and explain one patient




*Accuracy on patients the model never saw. Rules on one office reading or the daytime average are right 54% and 61% of the time. The gain over simpler machine learning is small and within the margin of error; the model's edge is better-calibrated confidence and explanations.*




*The dashboard's ABPM Staging page, captured from the running app: the predicted stage with its confidence, stage probabilities, and the four measurements that drove the result. Night-time pressure carries the most weight, a pattern built into the simulated data.*

**Where it errs.** Most mistakes fall within 5 mmHg of a stage boundary, where the stage is a judgment call. Accuracy is lowest under age 45 (86%, 93 patients), so it needs a larger test.

## 6 Limits and Next Steps

- **Simulated data only.** Not clinically validated and not for patient care.
- **Next:** test on real multi-site records, predict real outcomes, and show prediction sets with coverage guarantees near stage boundaries (Angelopoulos & Bates, 2023).
- **Planned, not built:** clinician override logging and retraining.

## References

Angelopoulos, A. N., & Bates, S. (2023). Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning, 16*(4), 494–591. https://doi.org/10.1561/2200000101

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